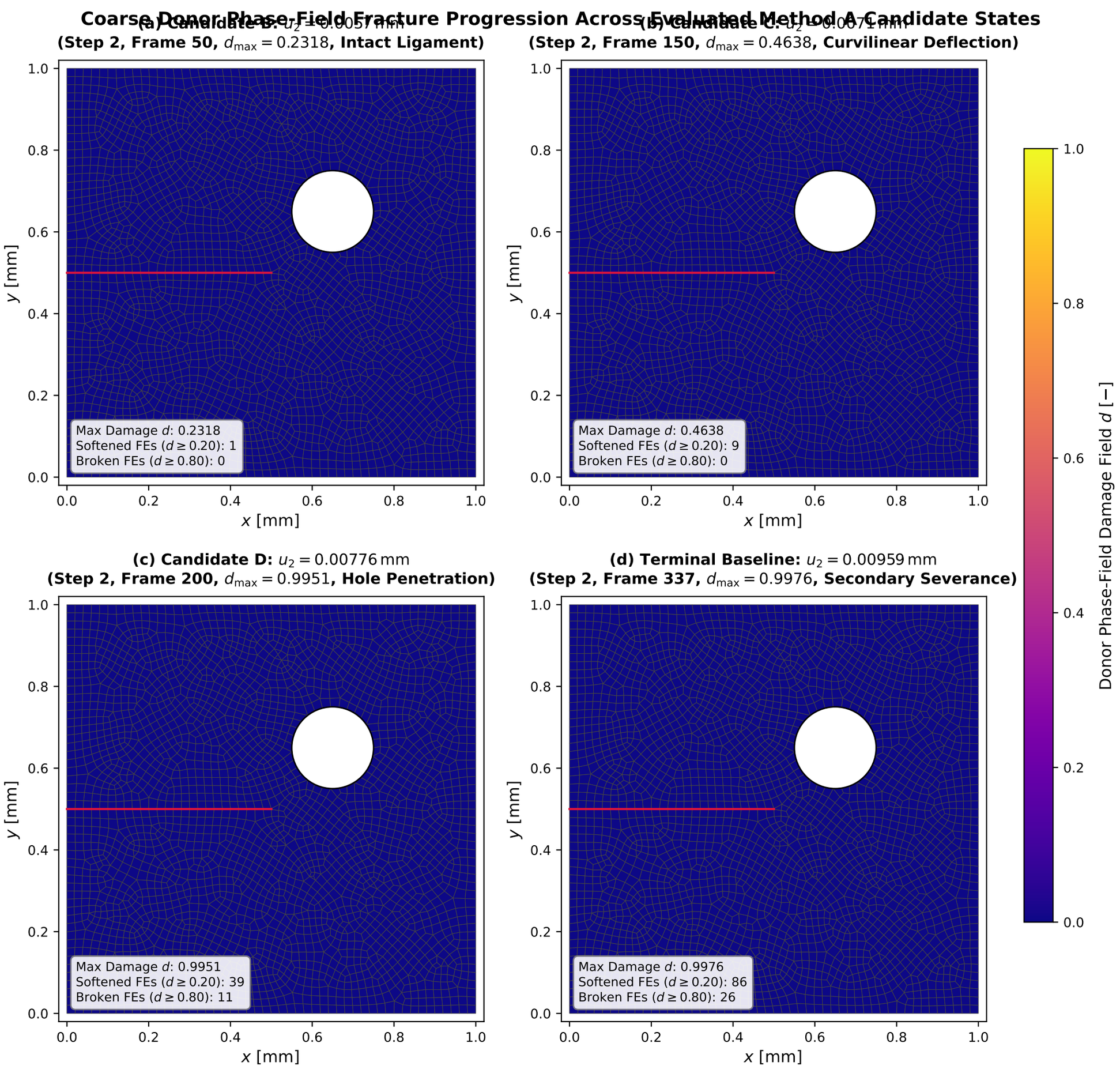

Coarse Donor Phase-Field Fracture Progression Across Evaluated Method A Candidate States
(a) Candidate B: $u_2 = 0.0057\,\mathrm{mm}$
(Step 2, Frame 50, $d_{\max} = 0.2318$, Intact Ligament)
Max Damage d: 0.2318
Softened FEs ($d \geq 0.20$): 1
Broken FEs ($d \geq 0.80$): 0
(b) Candidate C: $u_2 = 0.00771\,\mathrm{mm}$
(Step 2, Frame 150, $d_{\max} = 0.4638$, Curvilinear Deflection)
Max Damage d: 0.4638
Softened FEs ($d \geq 0.20$): 9
Broken FEs ($d \geq 0.80$): 0
(c) Candidate D: $u_2 = 0.00776\,\mathrm{mm}$
(Step 2, Frame 200, $d_{\max} = 0.9951$, Hole Penetration)
Max Damage d: 0.9951
Softened FEs ($d \geq 0.20$): 39
Broken FEs ($d \geq 0.80$): 11
(d) Terminal Baseline: $u_2 = 0.00959\,\mathrm{mm}$
(Step 2, Frame 337, $d_{\max} = 0.9976$, Secondary Severance)
Max Damage d: 0.9976
Softened FEs ($d \geq 0.20$): 86
Broken FEs ($d \geq 0.80$): 26
x [mm]
y [mm]
Donor Phase-Field Damage Field d [−]